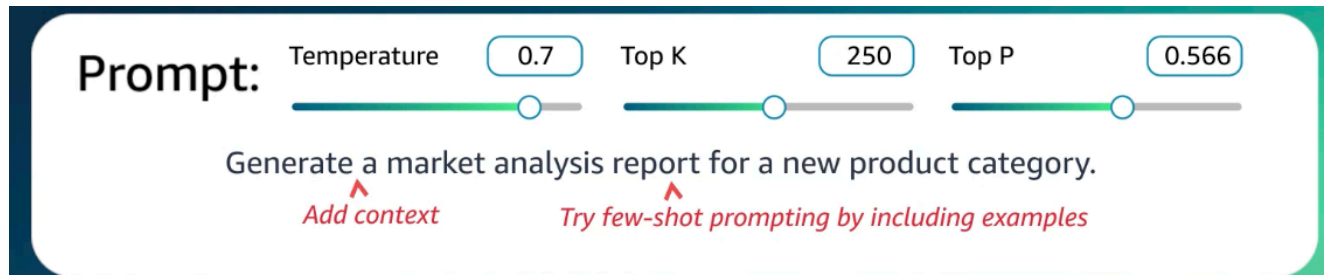


Essentials of Prompt Engineering



Using effective prompt strategies can offer you the following benefits:

- Enhance the model's capabilities and bolster its safety measures.
- Equip the model with domain-specific knowledge and external tools without modifying its parameters or undergoing fine-tuning.
- Interact with language models to fully comprehend their potential.
- Obtain higher-quality outputs by providing higher-quality inputs.

Elements of a prompt

- **Instructions:** This is a task for the large language model to do. It provides a task description or instruction for how the model should perform.
- **Context:** This is external information to guide the model.
- **Input data:** This is the input for which you want a response.
- **Output indicator:** This is the output type or format.

Negative prompting

Sometimes it's easier to guide a model toward a desired output by including what you don't want included in the output. Negative prompting is used to guide the model away from producing certain types of content or exhibiting specific behaviors. It involves providing the model with examples or instructions about what it should not generate or do.

For instance, in a text generation model, negative prompts could be used to prevent the model from producing hate speech, explicit content, or biased language. By specifying what the model should avoid, negative prompting helps steer the output towards more appropriate content.

Modify the prompts

Although foundation models (FMs) are generally highly capable, their outputs can be greatly influenced by the prompts provided. In this lesson, you will discover techniques for modifying and refining prompts to achieve better results. By the end of this lesson, you will have a solid understanding of how to tweak and optimize prompts, unlocking the full potential of generative AI models.

Inference parameters

When interacting with FMs, you can often configure inference parameters to limit or influence the model response. The parameters available to you will vary based on the model that you are using. Inference parameters fit into a range of categories, with the most common being randomness and diversity and length.

Randomness and Diversity

This is the most common category of inference parameter. Randomness and diversity parameters influence the variation in generated responses by limiting the outputs to more likely outcomes or by changing the shape of the probability distribution of outputs. Three of the more common parameters are temperature, top k, and top p. Choose each to learn more.

- **Temperature** = This parameter controls the randomness or creativity of the model's output. A higher temperature makes the output more diverse and unpredictable, and a lower temperature makes it more focused and predictable. Temperature is set between 0 and 1. The following are examples of different temperature settings.
- **Top p** is a setting that controls the diversity of the text by limiting the number of words that the model can choose from based on their probabilities. Top p is also set on a scale from 0 to 1. The following are examples of different top p settings.
 - With a low top p setting, like 0.250, the model will only consider words that make up the top 25 percent of the total probability distribution. This can help the output be more focused and coherent, because the model is limited to choosing from the most probable words given the context.
 - With a high top p setting, like 0.990, the model will consider a broad range of possible words for the next word in the sequence, because it will include words that make up the top 99 percent of the total probability distribution. This can lead to more diverse and creative output, because the model has a wider pool of words to choose from.
- **Top k** limits the number of words to the top k most probable words, regardless of their percent probabilities. For instance, if top k is set to 50, the model will only consider the 50 most likely words for the next word in the sequence, even if those 50 words only make up a small portion of the total probability distribution.
 - With a low setting, like 10, the model will only consider the 10 most probable words for the next word in the sequence. This can help the output be more focused and coherent, because the model is limited to choosing from the most probable words given the context.
 - With a high top k setting, like 500, the model will consider the 500 most probable words for the next word in the sequence, regardless of their individual probabilities. This can lead to more diverse and creative output, because the model has a larger pool of potential words to choose from.

Length

The length inference parameter category refers to the settings that control the maximum length of the generated output and specify the stop sequences that signal the end of the generation process. To learn more, choose each of the following parameters.

- **The maximum length** setting determines the maximum number of tokens that the model can generate during the inference process. This parameter helps to prevent the model from generating excessive or infinite output, which could lead to resource exhaustion or undesirable behavior. The appropriate value for this setting depends on the specific task and the desired output length. For instance, in natural language generation tasks like text summarization or translation, the maximum length can be set based on the typical length of the target text. In open-ended generation tasks, such as creative writing or dialogue systems, a higher maximum length might be desirable to allow for more extended outputs.
- **Stop sequences** are special tokens or sequences of tokens that signal the model to stop generating further output. When the model encounters a stop sequence during the inference process, it will terminate the generation regardless of the maximum length setting. Stop sequences are particularly useful in tasks where the desired output length is variable or difficult to predict in advance. For example, in conversational artificial intelligence (AI) systems, the stop sequence could be an end-of-conversation token or a specific phrase that indicates the end of the response.
Stop sequences can be predefined or dynamically generated based on the input or the generated output itself. In some cases, multiple stop sequences can be specified, allowing the model to stop generation upon encountering any of the defined sequences.

Best practices for prompting

- Be clear and concise
- Include the context if needed
- Use the directives for the appropriate response type
- Consider the output in the prompt
- Start prompts with an interrogation
- Provide an example response
- Break up the complex tasks
- Experiment and be creative
- Use prompt templates

Technique for Prompting

- Zero-shot prompting
- Few-shot prompting
- Chain-of-thought prompting = a technique that divides intricate reasoning tasks into smaller, intermediary steps. This approach can be employed using either zero-shot or few-shot prompting techniques. CoT prompts are tailored to specific problem types. To initiate the chain-of-thought reasoning process in a machine learning model, you can use

the phrase "Think step by step." It is recommended to use CoT prompting when the task requires multiple steps or a series of logical reasoning.

Prompt Misuse and Risks

- **Poisoning:** refers to the intentional introduction of malicious or biased data into the training dataset of a model. This can lead to the model producing biased, offensive, or harmful outputs, either intentionally or unintentionally.
- **Hijacking and prompt injection:** Hijacking and prompt injection refer to the technique of influencing the outputs of generative models by embedding specific instructions within the prompts themselves.

Exposure and the prompt leaking

Exposure = the risk of exposing sensitive or confidential information to a generative model during training or inference. An FM can then inadvertently reveal this sensitive data from their training corpus, leading to potential data leaks or privacy violations.

Jailbreaking